

SMART CAMPUS 360:

AI + IOT SMART CAMPUS INTELLIGENCE

TEAM NAME

APEX DEVS

TEAM MEMBERS

Ananya Bansal – AI/ML & Data Analytics Lead

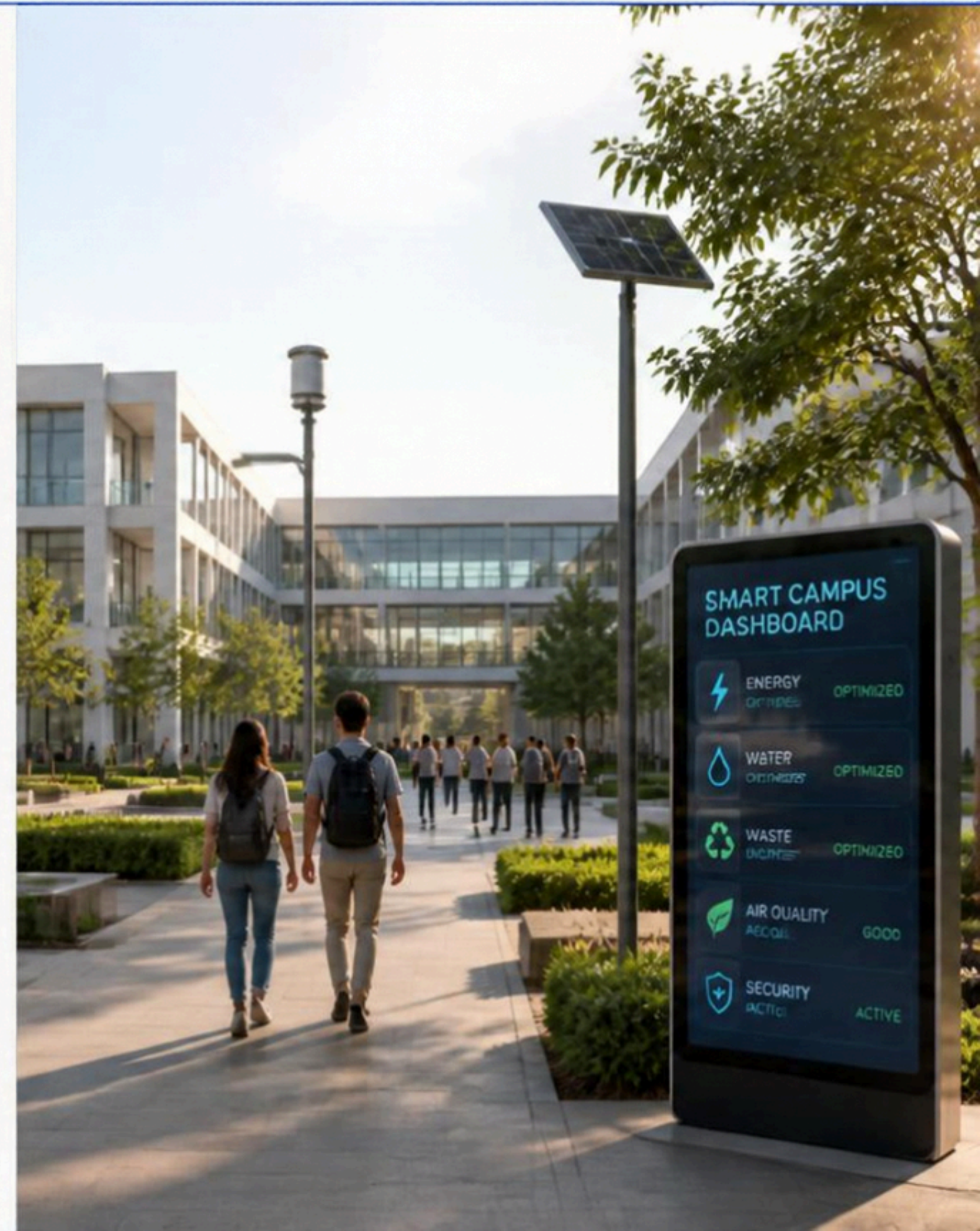
Arushi Shrikar – Frontend & Student Portal Lead

Akshay Singhal – Backend & Database Lead

Anurag Saini – IoT & System Integration Lead

SELECTED TRACKS

AI/ML Sustainability Internet of Things Open Innovation



PROBLEM STATEMENT & TARGET USERS

THE CHALLENGE

Campus data is fragmented across essential services
Issues are detected late and resources are underused
Communication gaps reduce awareness and engagement

WHO IT SERVES

Students, Faculty, Campus Administrators
Department Heads, Coordinators
Club Leaders, Event Organizers

An institutional platform with an accessible student interface



UNIQUE SOLUTION & CORE FEATURES

AI + IOT MONITORING

Electricity usage, smart bins, temperature & air quality, water meters

AI ANALYTICS

Consumption prediction, anomaly detection, waste classification, resource optimization

STUDENT PORTAL

Report issues, upload images, track complaints

ADMIN DASHBOARD

Live sensor data, alerts, complaint management, sustainability analytics

CONTACT LOGS

Verified photos & phone numbers for staff; privacy & role-based access

CLUB DIRECTORY

Club contacts, president, VP, senior members' year/class, club coordinator details

HACKATHON & EVENT DISCOVERY

Event info, dates, registration, contacts

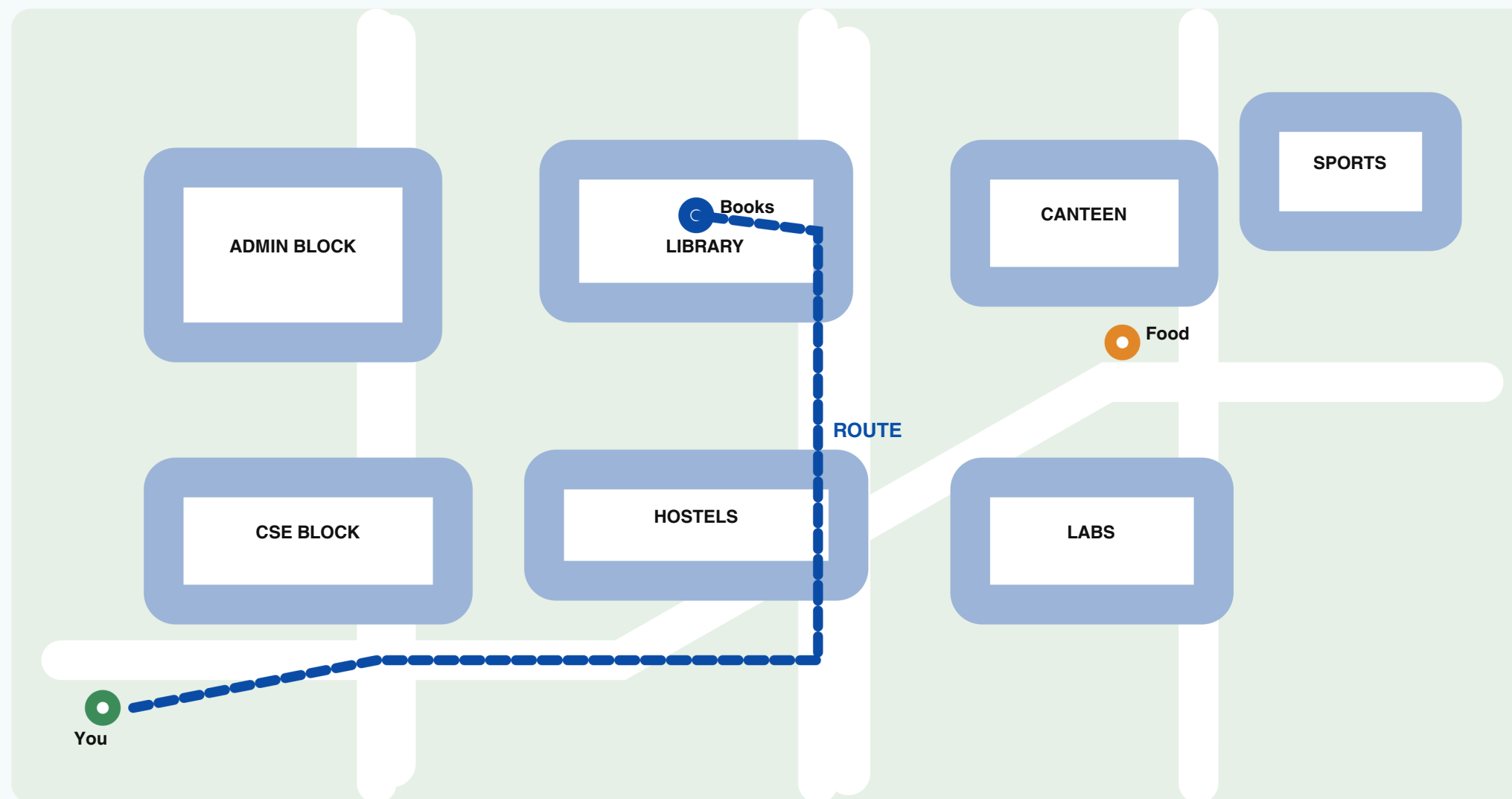


CAMPUS MAP & LOCAL INTELLIGENCE

A Google Maps-like experience built specifically for the college

INTERACTIVE WHOLE-CAMPUS MAP

Click any building, facility or food point to open its live information.



SMART ROUTING

- Multiple viable routes between any two campus points
- Shortest / accessible / landmark-based route options
- Walking distance, ETA, turn-by-turn guidance & live closures

FOOD & CANTEEN EXPLORER

- Click any canteen, café or food outlet on the map
- Live menu with every available food item and price
- Outlet timing, availability and location in one place

LIBRARY EXPLORER

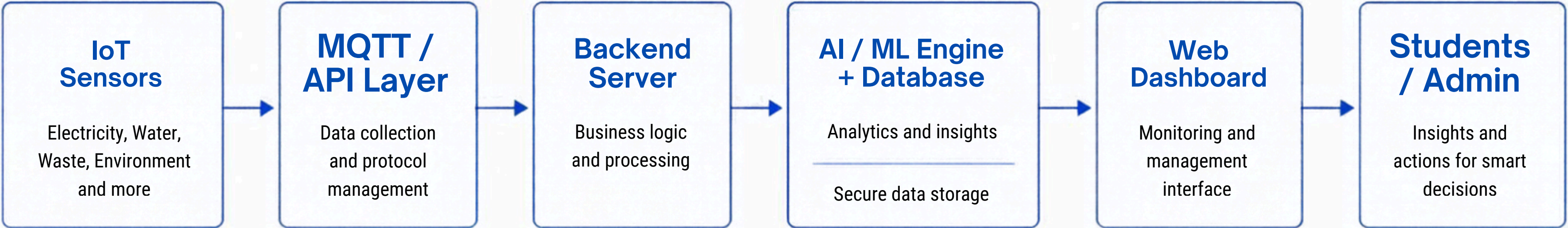
- Search the complete college book inventory by section
- Biotechnology • Programming • Competitive • Novels • Chemical Engg. • Shelf/section, availability and teacher-in-charge for borrowing

50 KM NEIGHBOURHOOD INTELLIGENCE

- Discover useful places within a maximum 50 km radius
- Hospitals, pharmacies, banks, food, transport, stores & PG's
- Map-based search with distance and route from campus

ONE MAP • EVERY CAMPUS SERVICE • LOCALITY AROUND IT

TECHNOLOGY STACK & ARCHITECTURE



KEY COMPONENTS

FRONTEND

React

BACKEND

Node.js / Express

DATABASE

PostgreSQL or MongoDB

MESSAGING

MQTT

AI / ML

Python, TensorFlow, scikit-learn

PLATFORM SERVICES

Cloud Deployment Image
Storage Authentication
Role-based Access

FEASIBILITY, RISKS, USP & BUSINESS MODEL

FEASIBILITY

Uses proven AI, IoT and web technologies

Centralized database supports campus information

Simple interface enables easy user access

Modular design allows future feature integration

Cloud deployment reduces infrastructure and maintenance costs

SHOWSTOPPERS & MITIGATION

Map data gaps → Use verified campus maps.

Sensor availability → Use simulated data initially.

Data accuracy → Add validation and admin review.

Large campus data → Use modular database architecture.

AI prediction errors → Keep human verification for alerts.

OUR USP

Unified platform connecting resource intelligence with campus navigation and student engagement

Proactive alerts plus verified contact, club and event discovery

BUSINESS MODEL

Subscription / annual institutional license

Implementation / integration fees

Optional analytics / support tiers



REFERENCES+ROADMAP+VIDEO

ROADMAP

01 — Foundation

Campus database + IoT integration
Maps, facilities, users and basic dashboard

02 — Intelligence

AI analytics + anomaly detection
Resource prediction, alerts and smart insights

03 — Student Layer

College Maps + Campus Connect
Routes, food menus, library and student services

04 — Smart Campus

Unified AI-powered campus platform
Sustainability, neighbourhood insights and smart decisions

REFERENCES

Official MQTT, React, Node.js, TensorFlow/scikit-learn docs

PostgreSQL/MongoDB, Cloud IoT platforms

Sustainability & smart-campus research papers

Open datasets and relevant case studies

OUR INNOVATION — CAMPUS CONNECT

A verified digital campus layer integrating:

- College Maps & multiple campus routes
- Food outlets, menus & prices
- Library books, sections & teachers
- Authorities, clubs, coordinators & events
- Hackathons and student opportunities
- 50 km neighbourhood information
- Institution-verified contacts and information

Integrated as a smart layer within SmartCampus 360.



PROJECT EXPLANATION VIDEO :

WATCH VIDEO